EXPERIMENT – 3 : Sampling, Quantization and Image Interpolation

Aim

To study the effects of reducing spatial resolution (sampling) and intensity resolution (quantization) on image quality, and to compare nearest-neighbour, bilinear and bicubic interpolation methods using MATLAB.

Software Required

- Personal computer
- MATLAB R2020a or later
- Image Processing Toolbox

Program

```
clc;
clear;
close all;
% Read grayscale image
I = imread('cameraman.tif');
%% Part A: Spatial Resolution (Sampling)
f = [1 2 4 8];
figure('Name','Sampling');
for k = 1:numel(f)
   Is = I(1:f(k):end, 1:f(k):end);
   Id = imresize(Is, size(I), 'nearest');
   subplot(2,2,k);
   imshow(Id);
   title(sprintf('%d x %d samples', ...
      size(Is,1), size(Is,2)));
end
%% Part B: Intensity Resolution (Quantization)
L = [256 64 16 8 4 2];
```

```
figure('Name','Quantization');
for k = 1:numel(L)
   step = 256 / L(k);
   Iq = uint8(floor(double(I)/step) * ...
      (255/(L(k)-1)));
   subplot(2,3,k);
   imshow(Iq);
   title(sprintf('%d levels (%d bits)', ...
      L(k), log2(L(k))));
end
%% Part C: Interpolation Methods
small = imresize(I, 0.25);
Nn = imresize(small, 4, 'nearest');
Bl = imresize(small, 4, 'bilinear');
Bc = imresize(small, 4, 'bicubic');
figure('Name','Interpolation');
subplot(2,2,1);
imshow(small);
title('64 x 64 Input');
subplot(2,2,2);
imshow(Nn);
title('Nearest Neighbour');
subplot(2,2,3);
imshow(Bl);
title('Bilinear');
subplot(2,2,4);
imshow(Bc);
title('Bicubic');
```

% Calculate PSNR

p1 = psnr(Nn, I);

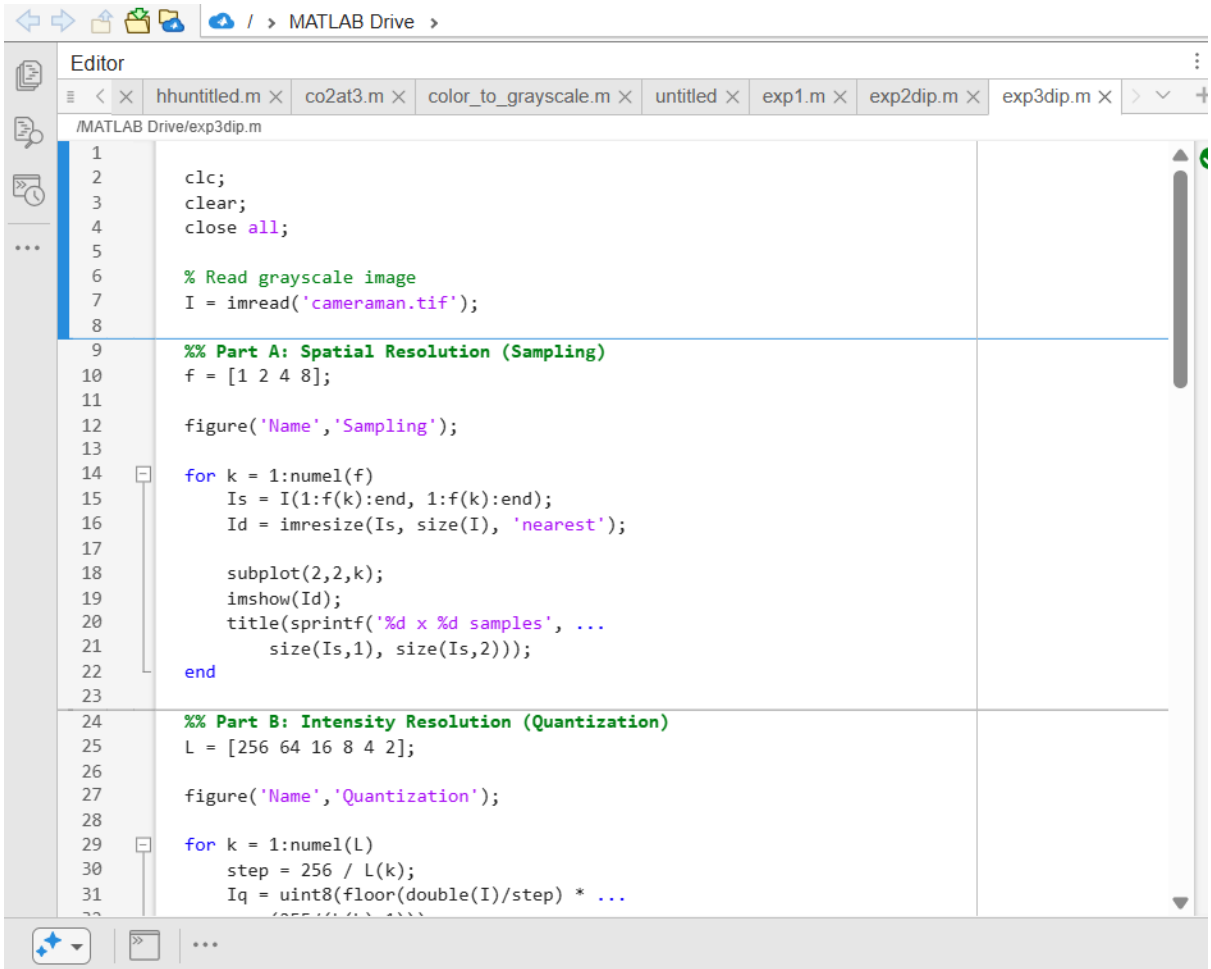
p2 = psnr(Bl, I);

p3 = psnr(Bc, I);

fprintf('PSNR Nearest Neighbour = %.2f dB\n', p1);

fprintf('PSNR Bilinear = %.2f dB\n', p2);

fprintf('PSNR Bicubic = %.2f dB\n', p3);



Procedure

1. Open MATLAB and set the Current Folder to your working directory.
2. Create a new script and save it as exp03.m.
3. Enter the program and run it by pressing F5.
4. Observe how reducing the number of samples affects spatial resolution.
5. Observe the changes in image quality as the number of gray levels decreases.

6. Compare nearest-neighbour, bilinear and bicubic interpolation methods.
7. Record the PSNR values displayed in the Command Window.
8. Repeat Part C using a shrink factor of 0.5 and compare the results.

Expected Output

Figures

Figure 3: Interpolation | Figure 2: Quantization | Figure 1: Sampling

Figure

256 x 256 samples



128 x 128 samples



64 x 64 samples



32 x 32 samples



Figures

Figure 3: Interpolation | Figure 2: Quantization | Figure 1: Sampling

256 levels (8 bits)



64 levels (6 bits)



16 levels (4 bits)



8 levels (3 bits)

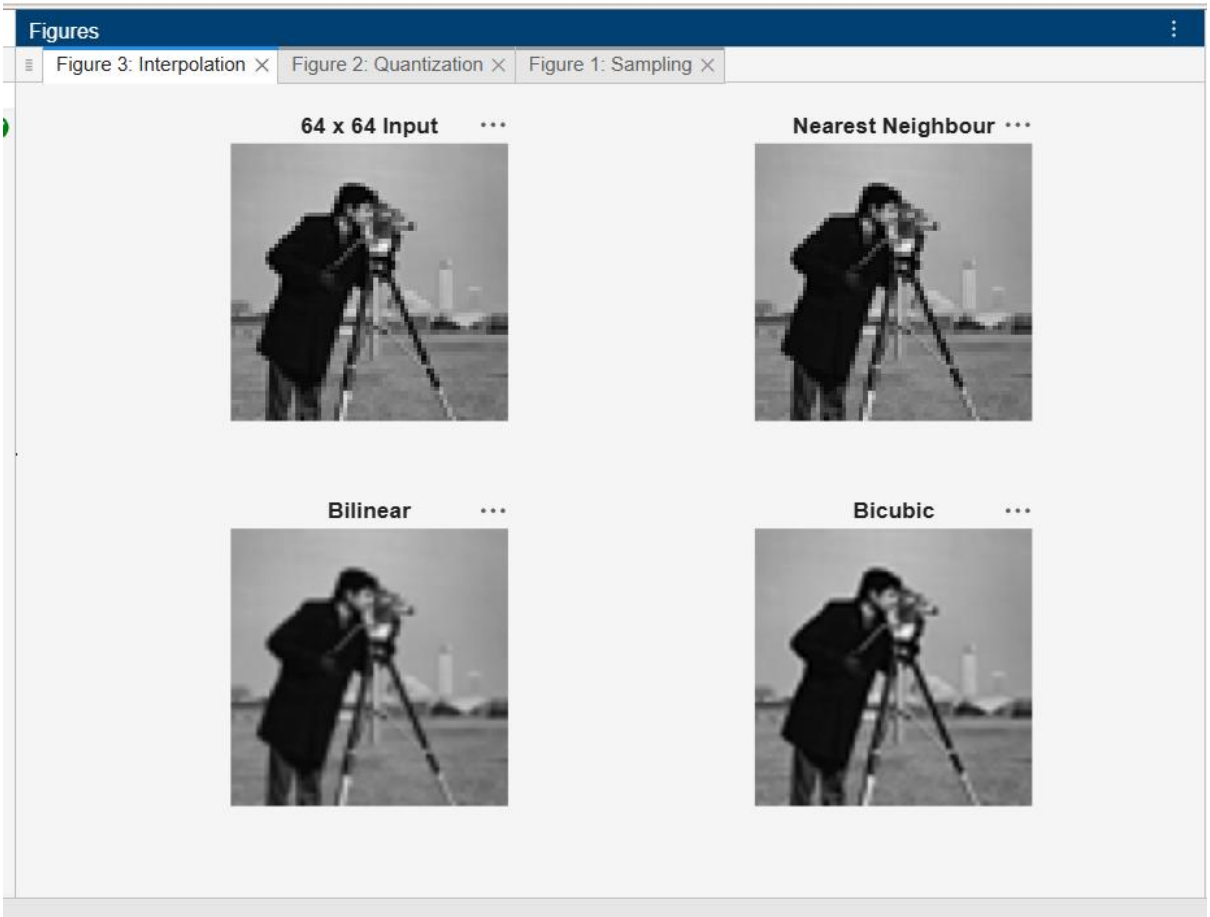


4 levels (2 bits)



2 levels (1 bits)





Observations

Interpolation method	PSNR (dB), factor 0.25	PSNR (dB), factor 0.5
Nearest neighbour	Record output	Record output
Bilinear	Record output	Record output
Bicubic	Record output	Record output

Note: Run the program to obtain the actual PSNR values.

Result

The effects of sampling and quantization on image quality were studied. Nearest-neighbour, bilinear and bicubic interpolation methods were compared using PSNR in MATLAB.